

Dhanesh Ramachandram, Ph.D

Sr.Member, IEEE, Applied Machine Learning Scientist

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Professional Profile

I am a seasoned machine learning researcher and applied scientist with a strong focus on healthcare AI: clinical dataset design, interpretable/explainable modeling, and translating research into deployed clinical systems. I have a proven track record in machine learning and medical image analysis research, having achieved a h-index of 17 with over 1900 citations to date. My expertise spans applied machine learning and deep learning for healthcare, including clinical dataset curation and evaluation, deep survival analysis for

competing clinical risks, interpretable/explainable AI (concept bottleneck models, neural additive models), LLM- and RAG-based clinical systems, and biomedical image analysis (semantic/instance segmentation, object detection, image classification). I have led clinical AI deployments at hospital sites, large-scale healthcare data annotation and extraction pipelines, and system-level interpretability research for agentic AI.

Experience

Vector Institute for AI

TORONTO, ON

Applied Machine Learning Scientist - Health Lead

Aug '23 – Present

- Designed and deployed **CRISPNAM-FG**, an intrinsically interpretable Fine-Gray competing-risks deep survival model, into production at St. Michael's Hospital to flag diabetic patients at high risk of post-discharge foot complications; built on a retrospective cohort of **107,000+ patients across 29 Ontario hospitals** (GEMINI database) and packaged the full deployment stack (Python wrapper around an R model, Flask UI, Docker containerization).
- Led an end-to-end **document data-extraction pipeline** (OLTCA project) combining Python PDF parsers (PyPDF2, pdfplumber, PyMuPDF) with LLM/OCR APIs (GPT-4, Google Document AI, DeepSeek OCR) to structure **37,000+** Ontario long-term-care inspection reports into a query-able dataset at $\geq 90\%$ target accuracy; delivered the codebase, searchable database, technical report and handoff.
- Led Vector's **Interpretability Bootcamp** (Apr–May 2025): organized 40+ participants across 12 teams, delivered lectures and hands-on mentorship, and produced 16 reference implementations of post-hoc and intrinsically interpretable models across tabular, NLP and imaging modalities; co-authored the resulting white paper (arXiv:2507.23535).
- Co-lead of Vector's **Interpretability Focus Area**; scoped and directed research projects on interpretable deep survival modeling and vision-language model interpretability, including concept bottleneck models for radiology grounded via retrieval-augmented citation and verification.
- Scientific advisor for an **LLM-based classification system** for long-context clinical conversations (CAMH), advising on deployment infrastructure, data security, licensing, labeling tooling and data readiness.
- Technical lead for an ultrasound lymph-node **detection/segmentation** project (HealthSpark:Tenomix): extended a 2D object detection model to a 2.5D volumetric pipeline and applied positive-unlabeled learning to address limited ground truth.
- Scientific advisor for a clinical **Text-to-SQL** system (HealthSpark:Verto Health): conducted cohort evaluation and literature review, and built a reference LightRAG-based implementation adopted by the delivery team.
- Scientific advisor for **surge classification and needle tracking** project (HealthSpark:Revolve Surgical): to build a model of action classification for surgical video and tracking needle-keypoints for robot-assisted surgery.
- Contributed statistical process control-based drift/monitoring methods to **Cyclops**, Vector's open-source healthcare ML monitoring toolkit.
- Technical lead on an **EHR comorbidity analytics** collaboration (CVD + T2D) with Boehringer-Ingelheim and Diabetes Action Canada, built a survival analysis model, resolving data and HPC infrastructure issues to enable model feature discovery.
- **Mentored 6+** research associates, interns and engineers on interpretable ML, survival analysis and LLM-based systems; supervised delivery of reference implementations and co-authored resulting papers/preprints.
- Reviewed healthcare AI deployments as technical consultant for the Sunnybrook Vector AI Initiative.

University of Guelph

GUELPH, ON

Adjunct Professor - School of Engineering

Aug '23 – Present

- Co-taught **UNIV*6990 AI and Society** (graduate course), developing and delivering lecture content on the AI-in-healthcare landscape across the US/Canada.
- Covered topics including development and deployment of AI in healthcare domains, bias and fairness in AI, and regulatory/governance aspects of AI deployment in healthcare.

- On-going University of Guelph-Vector Institute research collaborations for interpretable concept-bottleneck model for biodiversity (fine-grained species) classification.

Swift Medical Inc.

TORONTO, ON

Staff Machine Learning Scientist

Jan '18 – Aug '23

- Led the research and development of the earliest known commercial implementation of a deep learning-based **chronic wound segmentation model** for Swift Medical's Skin & Wound app and currently deployed to over 4,100 healthcare facilities in all states of the US and Canada. It is used on approx. 400,000 wound assessments per week.
- Led a project to annotate over 55,000 chronic wound tissue images and subsequently developed a mobile-friendly **wound tissue segmentation model** to aid clinicians objectively report tissue proportions in wound assessments.
- Implemented a deep learning model to detect Personal Health Information (PHI) in wound images ensuring HIPAA compliance for imaging data used for building AI models.
- Led a project to annotate over 100k Pressure Injury Wound images and implemented a **multitask deep learning model** for Pressure Injury Staging, wound depth and skin loss classification.
- Implemented a Mamdani-type **fuzzy inference system** for automatic Pressure Injury Staging system.
- Researched and implemented **self-supervised methods** for learning representations using a large unlabeled chronic wound images datasets for later fine-tuning for various downstream tasks.
- Implemented **real-time object detection model** for wound and fiducial detection enabling automatic wound image capture.
- **Formulated the research roadmap** for Clinical AI stream at Swift Medical and managed R&D projects within the stream.
- **Collaborative work with cross-functional teams** including infrastructure, product, engineering, clinical success teams to ensure successful integration of R&D outputs into products and monitoring model performance metrics post deployment.
- **Published core research findings** in peer-reviewed journals and wound care conferences.

NextAI

TORONTO, ON

Scientist In Residence

Mar '17 – Dec '17

- Provided **scientific mentoring** and support for machine learning and deep learning challenges for startups in the NextAI startup incubator program.
- One startup in particular that I provided support for, *IntuitiveAI*, won the best-startup award for the 2017 cohort group.

University of Guelph

GUELPH, ON

Research Fellow (Machine Learning Research Group)

Dec '16 – Nov '17

- Primary investigator: deep learning for wound and skin lesion segmentation and classification
- Developed a novel **skin lesion segmentation model** based on deep hyper-column descriptors leading to a journal publication and Best Paper Award at CVIS 2017 conference.
- Placed in the top-5 teams for the 2017 ISIC Skin Lesion Classification challenge.
- Mentored engineering undergraduates on the development of a web-based super-pixel segmentation tool.

University of Guelph

GUELPH, ON

Research Fellow (Machine Learning Research Group)

Jun '15 – Sep '16

- Research focused on multimodal deep learning involving video, audio and skeletal pose modalities.
- Conducted performance benchmarking and optimization for 3D-convolutional neural network-based multimodal deep learning architectures
- Wrote a **highly cited review paper on deep multimodal learning** with over 500 citations to date.
- Co-developed a novel **structure optimization for deep multimodal fusion networks** using Bayesian Optimization
- Implemented various CNN architectures using Theano, Lasagne and an early adopter of Tensorflow. Experience using the SharcNet GPU clusters.

Universiti Sains Malaysia

PENANG, MALAYSIA

Associate Professor

Sep '03 – Jun '15

- Conducted fundamental and applied research in the area of computer vision, medical image analysis and image understanding leading to numerous high-impact conference and peer-reviewed journal publications.
- Successfully secured and managed numerous grants to support research.
- Supervised student research at undergraduate and postgraduate level.
- Provided technical consultation for the university and industry on computer vision and image processing.
- Taught linear algebra, programming, research methodology and computer vision courses at undergraduate and postgraduate levels.

- Performed integration, testing and performance tuning of vision inspection systems for various back-end semiconductor assembly equipment.
- Provided on-site and e-mail technical support on customer issues related to vision-based inspection systems installed on Ismeca products
- Conducted systems buy-off and validation of accuracy and repeatability of vision-based inspection systems prior to product shipment.
- Conducted in-house and customer training particularly related to vision-based inspection systems.

Please refer to [my Google Scholar profile](#) for the complete list of my scientific publications.

Education

Universiti Sains Malaysia

PENANG, MALAYSIA

Doctor of Philosophy

1997–2003

Research focused on artificial neural networks, structured lighting and sensory perception for robotic manipulators.

Universiti Sains Malaysia

PENANG, MALAYSIA

B.Tech in Industrial Technology

1993 – 1997

Publications

Journal Articles

- [1] Sindhuja Chaduvula et al. "From Features to Actions: Explainability in Traditional and Agentic AI Systems". In: *arXiv preprint arXiv:2602.06841* (2026).
- [2] Robert D. J. Fraser et al. "A Retrospective Analysis of Deep Tissue Pressure Injury Prevalence and Incidence Using a Large-scale Wound Care Database in Long-term Care Settings Across North America". In: *Advances in Skin & Wound Care* (2026). NPIAP 2026 Annual Conference Research Abstracts. doi: [10.1097/asw.0000000000000488](https://doi.org/10.1097/asw.0000000000000488).
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- [4] Sneha Rao, Shaina Raza, and Dhanesh Ramachandram. "CaVe-VLM-CoT: An Interpretable Vision-Language Model Framework". In: *arXiv preprint arXiv:2606.18385* (2026).
- [5] Surain B Roberts et al. "Development, validation, and pilot implementation of a prognostic model to predict future foot complications among people with diabetes recently discharged from hospital". In: *Journal of Clinical Epidemiology* (2026), p. 112295. URL: <https://www.sciencedirect.com/science/article/pii/S0895435626001708>.
- [6] Judy Zhu et al. "Interpreting Agentic Systems: Beyond Model Explanations to System-Level Accountability". In: *arXiv preprint arXiv:2601.17168* (2026).
- [7] Sara Ketabi and Dhanesh Ramachandram. "Bridging Electronic Health Records and Clinical Texts: Contrastive Learning for Enhanced Clinical Tasks". In: *arXiv preprint arXiv:2505.17643* (2025).
- [8] Dhanesh Ramachandram et al. "Transparent AI: The Case for Interpretability and Explainability". In: *arXiv preprint arXiv:2507.23535* (2025).
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- [11] Dhanesh Ramachandram et al. "Bayesian optimization on graph-structured search spaces: Optimizing deep multimodal fusion architectures". In: *Neurocomputing* 298 (2018), pp. 80–89.
- [12] Dhanesh Ramachandram and Graham W. Taylor. "Deep Multimodal Learning: A Survey on Recent Advances and Trends". In: *IEEE Signal Processing Magazine* 34.6 (2017), pp. 96–108. doi: [10.1109/MSP.2017.2738401](https://doi.org/10.1109/MSP.2017.2738401).
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- [14] Ahmed Adel Abu-Shareha et al. "Multimodal Concept Fusion Using Semantic Closeness for Image Concept Disambiguation". In: *Journal of Multimedia Tools and Applications* 61.1 (2012), pp. 69–86.

- [15] Mohamad Ehsan Abbasnejad, Dhanesh Ramachandram, and Rajeswari Mandava. "A Survey of the State of the Art in Learning the Kernels". In: *Knowledge and Information Systems* 31.2 (2012), pp. 193–221.
- [16] Kok Haur Ong et al. "Automatic White Matter Lesion Segmentation using an Adaptive Outlier Detection Method". In: *Magnetic Resonance Imaging* 30.6 (2012), pp. 807–823.
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- [22] Dhanesh Ramachandram and Rajeswari Mandava. "Neural Network based Robot Visual Positioning for Intelligent Assembly". In: *Journal of Intelligent Manufacturing* 15.2 (2004), pp. 219–231.

Books Chapters

- [1] Abdulqadeer Mohsin, Ahamad Tajudin Kader, and Dhanesh Ramachandram. "Recent Advances in Harmony Search". In: ed. by Z.W.Geem. Vol. 270. *Studies in Computational Intelligence*. Springer-Verlag, 2010. Chap. An Optimization Algorithm based on Harmony Search for RNA Secondary Structure Prediction, pp. 163–174.
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Conference Proceedings

- [1] Judy Zhu et al. "Position: Accountable Deployment of Agentic AI Demands Layered, System-Level Interpretability". In: *Forty-third International Conference on Machine Learning Position Paper Track*. 2026. URL: <https://openreview.net/forum?id=liNx7tarrT>.
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Technical Reports

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References

- 1. **Prof Graham W. Taylor**, *Professor and Canada Research Chair in Machine Learning*, School of Engineering, University of Guelph, Guelph, ON N1G2W1.
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